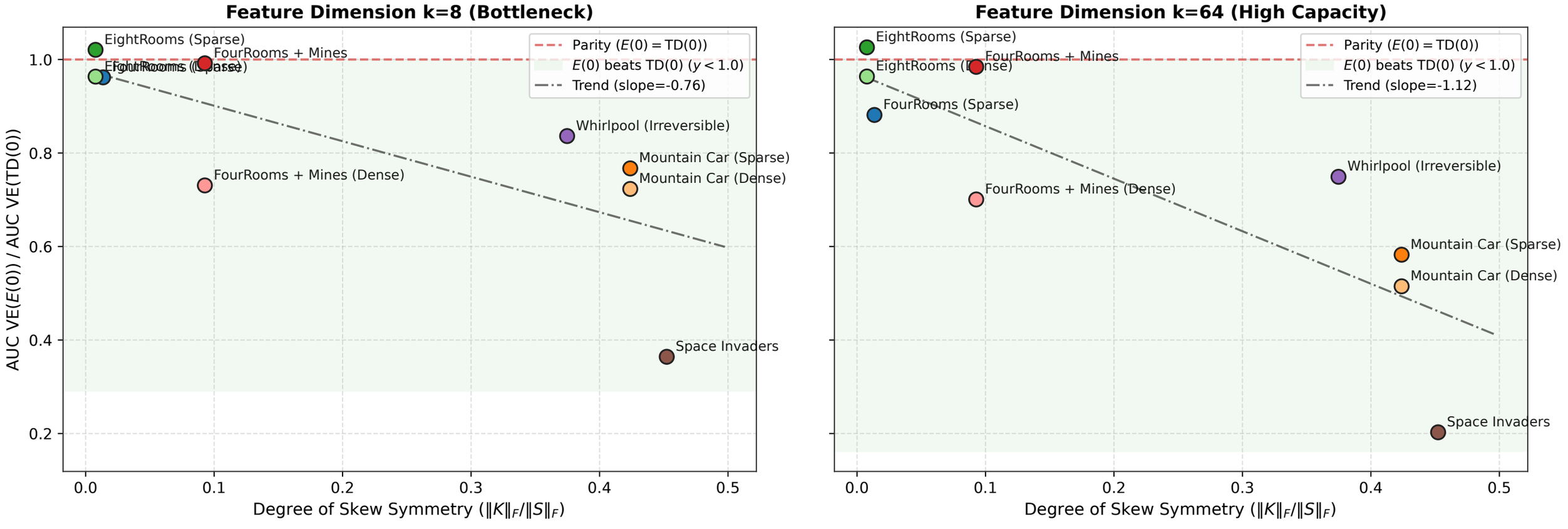
Impact of Markov Chain Skew-Symmetry on $E(0)$ vs. TD(0) Performance
Feature Dimension k=8 (Bottleneck)
Feature Dimension k=64 (High Capacity)
AUC VE($E(0)$) / AUC VE(TD(0))
Degree of Skew Symmetry ($\|K\|_F/\|S\|_F$)
Parity ($E(0) = \mathrm{TD}(0)$)
$E(0)$ beats TD(0) ($y < 1.0$)
Trend (slope=-0.76)
Trend (slope=-1.12)
EightRooms (Sparse)
FourRooms + Mines
FourRooms (Sparse)
Whirlpool (Irreversible)
Mountain Car (Sparse)
Mountain Car (Dense)
FourRooms + Mines (Dense)
Space Invaders
1.0
0.8
0.6
0.4
0.2
0.0
0.1
0.2
0.3
0.4
0.5